

Fault and stop codes

Read `FAULT_DETAIL` and `STOP_REASON` from the dashboard (`DASHBOARD.bat`) or with `foc_diag.py`. STS gives the coarse state; `FAULT_DETAIL` says **why**.

Codes come from `drive/hs300_drive.h`. If a code here disagrees with the firmware, the firmware is right — say so and this file gets fixed.

Drive state — STS

value	state	meaning
0	OFF	stopped, gates off
1	ON	running
2	FAULT	stopped by a fault; read <code>FAULT_DETAIL</code>
3	FAULT_OC	over-current
4	FAULT_OV	over-voltage
5	FAULT_UV	under-voltage
6	FAULT_UV_WAIT	waiting for the bus to come back
7	FAULT_OT	over-temperature

MODE: 0 = MANUAL, 1 = MPPT, 2 = DCBUS.

Over-current — 20 to 23

All four mean current, but not the same current, and the action differs.

code	name	what tripped	first thing to check
20	OC_ENVELOPE	id^2+iq^2 over the envelope	a real overload — pump jammed, sand, a dry well
21	OC_TRACKING	the current loop lost authority	the motor is not following: wrong tune, a weak bus, a failing phase
22	OC_RMS_TOC	RMS time-overcurrent, the thermal one	sustained overload rather than a spike. Check F01 against the plate
23	OC_INST	instantaneous per-phase	a short or a shoot-through. Do not just restart — look for a fault in the power stage or the motor leads

23 is the serious one. 20 and 22 usually mean the load; 23 usually means hardware.

Insulation and winding — 30 to 32

code	name	meaning	action
30	INSUL_LEAK	zero-sequence current — current going to earth	a wet joint or a damaged cable, 70 m down. Megger the motor and the cable before restarting
31	WINDING_ASYM	the three line-to-line resistances disagree	see the warning below before believing it

code	name	meaning	action
32	PHASE_OPEN	one lead broken, the other two intact	PS_FLAGS names which phase. Check the joints at the head first

31 is often not what it says

Read PS_FLAGS before trusting a 31. If bit 0x4000 (PS_F_OVERCUR) is set, a probe vector aborted on over-current and left *no measurement behind* — that is not asymmetry at all. Fix the over-current cause and re-run.

A relay or an ADC-offset problem can also fake an open phase. That exact chain cost a day on the grid board on 2026-09-19.

Failed auto-tune — 40 to 48

40 + failcode. The drive is stopped and **not tuned**; it will not run closed loop until a tune succeeds.

code	meaning	usual cause
41	Rs measurement failed	no motor, an open phase, or current sensing wrong
42	the 40 Hz injection failed	could not drive enough current — weak bus, or F01 too low
43	leakage inductance out of band	wrong plate, or the measurement is being clipped
44	the rotor did not turn	the most common. See below
45	the rotor dragged	something mechanical, or far too little bus
46	could not hold current	the bus sagged during the tune — tune on a stiff supply
47	the result disagrees with the nameplate	the plate is wrong, or it is not the motor you think
48	current abort	hit the limit; F01 or the plate is wrong

44 — the rotor did not turn

The gate behind 44 is derived from the **plate**, as $566 / \text{plate_I_rms}$ mH. So a plate current entered **too low RAISES the gate** and refuses a perfectly good spin. Check the plate before touching anything else.

On the 22 kW well machine PLATE_I_RMS is deliberately **45 A**, not the 39 A nameplate — that is intentional and documented. Do not "correct" it.

Tune on the stiffest supply you have. On PV at dawn there is not enough power and you will chase a fault that is really a lack of watts.

Clean stops that are not faults — 11 to 13

code	meaning
11	a STOP arrived over the serial link (a tool or the dashboard)
12	START_DEFER — the bus could not carry the start, so the drive did not fight it
13	LINGER — it was running below pump cut-in on a falling sun and gave up

12 and 13 are correct behaviour, not failures. They replaced a dawn pattern of ten UV trips in a row: the I-f start needs roughly 750 W of magnetising current and the array had 50–800 W. Probing and deferring is the fix.

Pre-start check — PS_FLAGS bits

Run before every start unless PRESTART_EN is 0.

bit	name	severity
0x01	SKIPPED	disabled by PRESTART_EN
0x02	NO_MOTOR	warn — all three pairs open, nothing connected
0x04	R_HIGH	warn — symmetric but high against the stored Rs
0x08	LEAK	fail — zero-sequence current
0x10	ASYM	fail — spread over the limit
0x20	OPEN	fail — one or two pairs open
0x40 / 0x80 / 0x100	PH_U / PH_V / PH_W	which phase the open pairs share
0x200 / 0x400 / 0x800	RAIL_UV / RAIL_VW / RAIL_WU	that vector railed the ADC
0x4000	OVERCUR	a vector aborted on over-current — no measurement was taken

PS_STATE: 7 = pass, 8 = failed. PS_VCODE packs four bits per line-to-line vector; 5 means that vector hit over-current.

A start within about 2 seconds of a stop fails the pre-start on a coasting motor — the back-EMF looks like a fault. Wait, then start.